GNBF5020

Lecture 6 · Exam Focus & Learn-from-Zero Guide

Methods in Molecular Biology — the named question, the three traps, the numbers, the pairs, and ten concept chains that take you from zero.

GNBF5020 · Introduction to Molecular Biology and Genetics

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Lecture No. 6 — Methods in Molecular Biology · 2 h 18 min · 7 Nov 2026

Assessment weighting (published): Assignments 30% · Mid-term 30% · Final 40%

Source: the 2 h 18 min recording and its 46-slide deck, transcribed in full, section by section

[ENGLISH ONLY] ALL EXAM ANSWERS MUST BE IN ENGLISH. Dr. Wong's own words, from Lecture 1: *"every assignments and exam should be done in English, even this is a Chinese University of Hong Kong."* And again in Lecture 4: *"you have to answer in English, although this is a CUHK, haha, okay."* He said he gave zero marks to students who answered in Chinese last year. In the Traditional Chinese edition of this book every warning box is in Chinese, but **all gene, protein, pathway, method and drug names stay in English**.

[HOW TO USE THIS BOOK] Two halves, two jobs. Part A is the exam half: what Dr. Wong actually said about the exam, the three traps he named, the numbers and thresholds, the seventeen discriminator pairs, the method-selection table, and a fifteen-question self-test with a full answer key — read it the night before the paper. Part B is the learning half: ten concept chains that take you from zero molecular biology to being able to *explain* the lecture rather than recognise it — work through it in order, with a pen in hand. Five figure plates summarise the sample frame, the named question, the two platform comparisons and the five-day plan.

繁體中文版 Traditional Chinese edition: GNBF5020_第六課考試重點與學習指南_繁體中文.pdf

Video guides

Twelve terms in this book carry the most exam weight, and each one has a short video guide: **English narration, Traditional Chinese subtitles**, about a minute each. The MP4 files live in the project folder **_video/assets/** - the full path is printed next to every link below, so you can still find a file after this PDF has been moved. **_video/index_lec6.html** plays all twelve on one page.

Video	Term	Chapter	Video file
L01	**Garbage in, garbage out**	Lecture 6 · Methods in Molecular Biology	_video/assets/L01_Garbage_in_garbage_out.mp4
L02	**FFPE vs fresh frozen tissue**	Lecture 6 · Methods in Molecular Biology	_video/assets/L02_FFPE_vs_fresh_frozen_tissue.mp4
L03	**Serum vs plasma**	Lecture 6 · Methods in Molecular Biology	_video/assets/L03_Serum_vs_plasma.mp4
L04	**SYBR Green vs TaqMan**	Lecture 6 · Methods in Molecular Biology	_video/assets/L04_SYBR_Green_vs_TaqMan.mp4
L05	**Microarray vs RNA-Seq**	Lecture 6 · Methods in Molecular Biology	_video/assets/L05_Microarray_vs_RNA_Seq.mp4
L06	**Bulk vs single-cell RNA-Seq**	Lecture 6 · Methods in Molecular Biology	_video/assets/L06_Bulk_vs_single_cell_RNA_Seq.mp4
L07	**ChIP-Seq**	Lecture 6 · Methods in Molecular Biology	_video/assets/L07_ChIP_Seq.mp4
L08	**Clonal expansion and the cancer vaccine**	Lecture 6 · Methods in Molecular Biology	_video/assets/L08_Clonal_expansion_and_the_cancer_vaccine.mp4
L09	**Reducing vs non-reducing SDS-PAGE**	Lecture 6 · Methods in Molecular Biology	_video/assets/L09_Reducing_vs_non_reducing_SDS_PAGE.mp4
L10	**Isoelectric focusing and 2D-PAGE**	Lecture 6 · Methods in Molecular Biology	_video/assets/L10_Isoelectric_focusing_and_2D_PAGE.mp4
L11	**LC-MS vs LC-MS/MS**	Lecture 6 · Methods in Molecular Biology	_video/assets/L11_LC_MS_vs_LC_MS_MS.mp4
L12	**Positive vs negative CRISPR screen**	Lecture 6 · Methods in Molecular Biology	_video/assets/L12_Positive_vs_negative_CRISPR_screen.mp4

Open the video index page ->

§0 How to use this book

This book does two jobs at once, and they are deliberately interleaved.

Part A is the exam half. It is built from what Dr. Wong actually said about the exam, not from a general sense of what might be examinable. Where he named a question, named a trap, or told you what the marks are for, it is quoted. Where he did not, it is left out. You should be able to read Part A the night before the paper and walk in with a short list.

Part B is the learning half. It assumes you are starting from zero on molecular biology methods and takes you to being able to *explain* the lecture rather than recognise it. It is organised as ten concept chains. A chain is not a list of facts; it is a line of reasoning you can reproduce out loud under exam pressure.

The two halves answer different questions, and you should read them differently:

	Part A — Exam Focus	Part B — Learn from Zero
Question it answers	*What will be on the paper?*	*How do I understand this at all?*
Read it by	reading, then drilling from memory	working through, in order, with pen in hand
Success looks like	you can recite the numbers and the pairs	you can explain why each technique exists
Time needed	3–4 hours total, in pieces	8–12 hours, spread over 5 days

One warning before you start. Dr. Wong's own remark at the top of the lecture sets the standard for everything in this book:

> *"Use simple sentences, short but clear sentence to make the description. That's the key to the exam."*

The marks are not for vocabulary. They are for a clear, mechanistically correct, short answer written in English. Every section below is written to be imitated in that style.

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§1 The lecture on one page

The lecture is built as **six moves**, each one narrowing the question until you are looking at a single cell, a single protein, or a single nucleotide. It is a funnel, and the funnel is the point.

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1 THE SAMPLE decides everything slide 3–6 "Garbage in, garbage out"
FFPE vs fresh frozen · serum vs plasma · central lab · dry run
↓
2 NUCLEIC ACIDS → cDNA → PCR slide 7–10 real-time PCR, SYBR vs TaqMan, multiplex
↓
3 MANY GENES AT ONCE slide 11–26 microarray vs RNA-Seq vs scRNA-Seq
↓
4 WHERE AND WHAT IS THE TUMOUR ANTIGEN slide 16–28 ChIP-Seq · clonal expansion · cancer vaccine
↓
5 PROTEINS: separate, then identify slide 29–36 SDS-PAGE · IEF · 2D-PAGE · mass spectrometry
↓
6 PROTEIN IN CELLS: perturb and express slide 37–45 CRISPR screen · transfection · fusion tags
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Three things are worth noticing about that shape.

First, **the sample comes before the technique.** Dr. Wong called the sample slide *"the most important slide of today classes"* and put it first, before any method. That is unusual and it is deliberate: in every later block he keeps returning to it (the degraded FFPE Bioanalyzer trace, the central-lab variation, the dry run).

Second, **each block answers a different question, and the question is the right way to tell the techniques apart.** Microarray, RNA-Seq and scRNA-Seq are not three versions of one thing; they answer *which genes are on?*, *how much of*

each?, and *which cell type is doing it?* respectively. If you learn the techniques as a list you will confuse them in the exam. If you learn them as answers to questions, you will not.

Third, **the last block is the experimental one.** CRISPR screening and recombinant protein expression are the two places where the lecture leaves "reading about" and starts "designing an experiment". These are the most likely to carry a scenario question.

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§2 The frame for everything: sample quality

Dr. Wong opened with the midpoint results and then told the class what the real problem was going to be. He also told you, in as many words, which slide matters most:

> *"This is the most important slide of today classes… I think for all the technique one day they will become obsolete. Some technique will be replaced by the others, new technique. But the principle is if you want to generate a data of high quality, the key is to have a high quality biological sample."*

> *"So if you are from the IT or computer science, you definitely know this jargon, right? Garbage in and garbage out… if you cannot use a quality sample, you definitely cannot generate a data with a high quality."*

This is the frame. Before you can ask "which technique", you must ask "is my sample even capable of answering the question". The lecture is full of places where the answer is *no*.

2.1 The three sample decisions

Decision	The options	What decides it
Tissue form	FFPE (formalin-fixed, paraffin-embedded) vs fresh frozen (FF)	FFPE is for histology and immunohistochemistry — the pathologist needs the architecture. For genomic profiling it is usually bad: the nucleic acid extracted from FFPE is degraded. **Use fresh frozen for gene-expression work.**
Blood matrix	Serum vs plasma	Serum = blood collected into a tube **without** anticoagulant, left at room temperature to clot, supernatant taken. Plasma = blood collected into a tube **with** anticoagulant (e.g. EDTA), centrifuged, supernatant taken. They are not interchangeable.
Homogeneity & handling	Bulk tissue vs isolated cells; tube type; tube orientation; shipment	A solid tumour is a mixture of tumour cells, normal cells and infiltrating immune cells. Bulk profiling averages them together and the signal of your target cell type is masked.

2.2 The evidence he put on screen

The rheumatoid arthritis biomarker. The study was aiming at early detection of RA by measuring a target DNA level in blood from RA patients versus healthy controls. Look at the plasma panel: **no difference at all**. Look at the serum panel: **a significant elevation in the RA patients**. Same target, same patients, same assay — a completely different result depending on the matrix. He attributes the difference to half-life and stability of the analyte in serum versus plasma.

If you are given an exam question that mentions a blood biomarker and does *not* say which matrix, the correct first sentence of your answer is that the matrix must be stated, because it can invert the result.

The Bioanalyzer trace. Human RNA has two dominant peaks, **28S and 18S** rRNA. Fresh-frozen tissue gives two sharp peaks. FFPE tissue gives a smear with no clean peak — degraded. That trace is the quality gate: it tells you whether the extraction worked before you spend money on sequencing.

The logistics. This is the part students underestimate and the part that is examinable. Dr. Wong's own group ran a Phase 1b trial across **Hong Kong, Shanghai and Australia** for a chronic hepatitis B biomarker, collecting **PBMC** from four Chinese cities — Guangzhou, Chengdu, Shanghai, Beijing — and shipping them to a single central lab in Shanghai for flow cytometry and NGS.

Three points came out of it:

1	**Central lab.** One lab analyses all samples from all sites, because "if you perform [the assay] by different people on a different day or even from a different centre, there must be a variation between the result". You validate the assay across operators and times so it is robust.

2	**Dry run.** Before collecting real samples you ship *fake* samples under different conditions to find out which conditions damage the sample. The readout is PBMC **viability**.
3	**Regulatory reality.** Biological samples cannot simply be shipped out of China. His solution was a central lab in Singapore for Hong Kong and Australia, and a *second* central lab in Shanghai for Chinese samples. That doubles the cost — which is a real trial cost, not an aside.

There is also a small, very testable detail: **the anticoagulant in the tube can poison the assay.** EDTA inhibits the enzyme in an ELISA readout and therefore *reduces* the signal. The colour you measure is partly an artefact of the tube you used.

And one result you should be able to argue about: in the dry run, shipping the blood tube **lying horizontal on a foam board beat shipping it vertical**, because more gas exchanges with the blood and the cells do not settle and pack against the bottom. It is a trivial finding and it is a perfect short-answer illustration of *why* you do a dry run.

[THE MARK-WINNING LINE] "A good sample is a precondition for a good dataset: no downstream technique can recover information that was destroyed in the sample — the quality of the profiling data is limited by the quality of the input material."

2.3 Self-check

1	A pathologist gives you a tumour block. You need to profile gene expression. What do you ask for, and why? → *Fresh frozen tissue, because FFPE degrades the nucleic acid; the Bioanalyzer 28S/18S trace is the QC gate.*
2	Your ELISA signal is lower in plasma samples than in serum samples of the same individuals. Before blaming the antibody, what do you check? → *The anticoagulant. EDTA inhibits the reporter enzyme and lowers the readout.*
3	Why does he insist on a central lab rather than each site running the assay locally? → *To remove operator, day and site variability so the data is comparable and the assay can be validated.*

Figure L1 — The sample gate: what decides whether a sample can be profiled at all

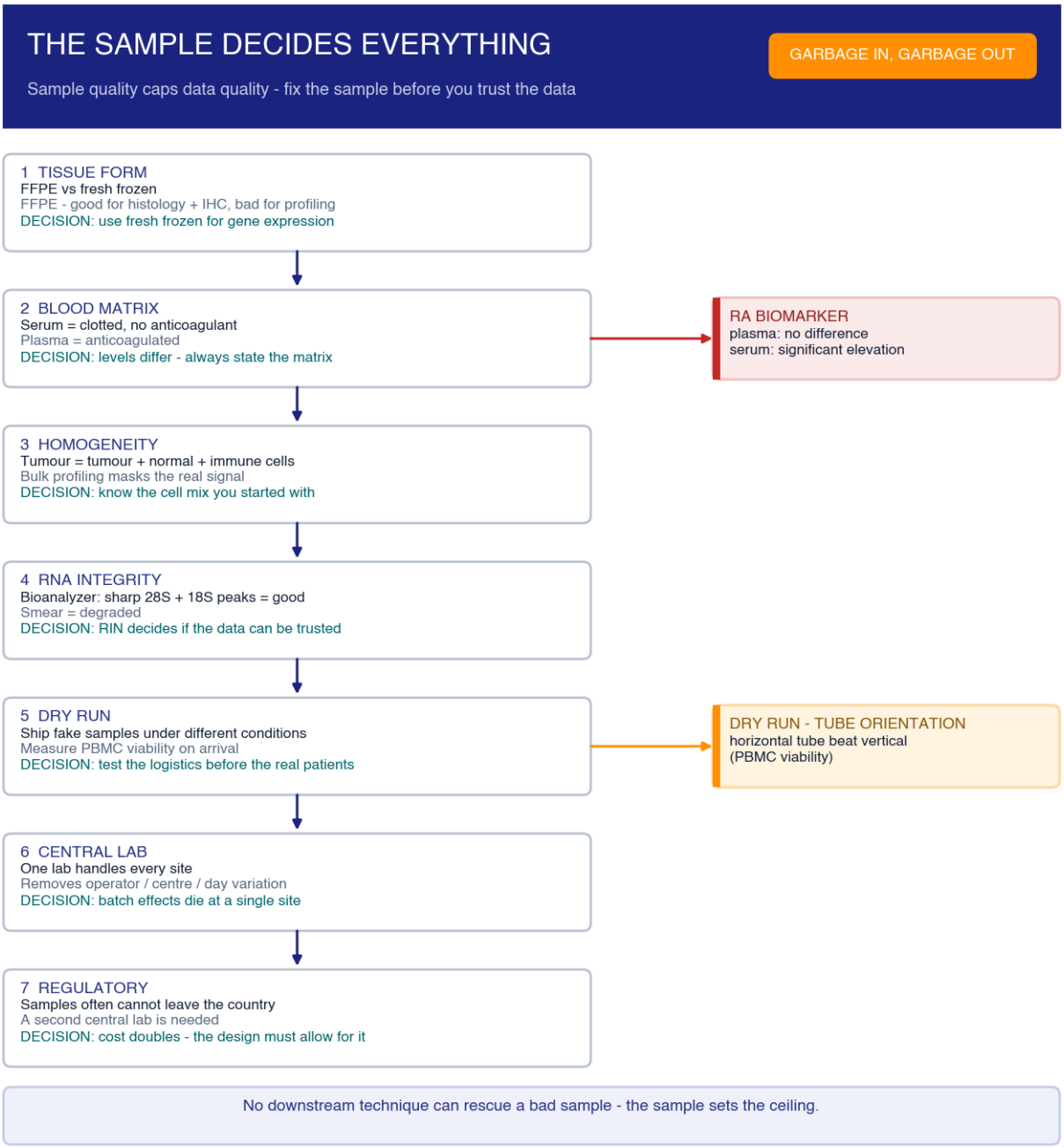


Figure L1. The sample decisions that come before any technique: tissue form, blood matrix, homogeneity, transport conditions, and the single central lab. Figure authored at 6.5 in print width and placed 1:1 on this page.

How to read this plate: work down the rows, because they are in the order the decisions are actually made. Every row is a gate - if a row goes wrong, every result underneath it is compromised. That is the 'garbage in, garbage out' argument, and it is why the sample frame comes before the techniques.

PART A — EXAM FOCUS

What Dr. Wong actually said about the exam: the named question, the three traps, the numbers, the seventeen pairs, the answer skeleton, and what he told you not to memorise.

A1 What he actually said about the exam

These are his words, in order of how much risk they carry for the final paper.

A1.1 The one he flagged by name ⚑

> *"Pay attention to this, because every year's most — not most, many student find very difficult to digest this positive versus negative screen."*

This is the highest-confidence prediction in the entire lecture. He singled out the **positive versus negative CRISPR screen** as something students repeatedly fail to digest, in the same way he used to single out the Kaplan–Meier plot in earlier lectures. Treat it as a named question.

A1.2 The pattern of failure he diagnosed ⚑

> *"Most student, especially for the question number three, so you basically you only describe the observation, but you don't show — you don't explain the phenomenon. Because the question I asked you to explain the survival differences between two patient groups… But I think many of you just give me a very lengthy, but not so focused description of that experimental data. So I think this is the one that you guys have to improve in your final exam."*

This is the single most useful thing he said. Read it twice:

- The question verb was **explain**. The answer was a **description**.
- The failure mode was not ignorance — it was **verb confusion**.
- The correction is not *know more* — it is to **name the mechanism that produces the observation**.

He repeated the same philosophy in the break conversation: *"you don't need to memorize the actual fact, but based on the basic principle, you know how to analyse the data."* And: *"just like the midterm, so you can understand that not emphasize on the details."*

A1.3 The trap in the immune-deficiency question ⚑

> *"A few student answered, saying the mismatch repair system is account for that immunodeficiency. Yeah, so this is not correct… Because the non-homologous end joining is responsible for the V(D)J recombination so that we can have a large population of T cell receptor and also the antibody."*

> *"I think some student only mentioned T cell receptor, or only the B cell for the antibody production. But the complete picture is the T cell receptor and also the antibody from the B cell or from the plasma cell."*

Two lessons. First, **NHEJ is the obligate pathway for V(D)J recombination; MMR is not involved.** Second, "partial credit" means naming only one limb of the answer: a complete answer needs **both** the TCR (Thymus) and the antibody (B cell / plasma cell) — one from each arm of adaptive immunity.

A1.4 The language rule ⚑

> *"Try to improve how you understand the questions and answer in English… I think the key is to improve your English by reading more research article… Use simple sentences, short but clear sentence to make the description. That's the key to the exam."*

He also noted the writing half of the problem: *"for some writing is very hard to understand."* Read primary literature — that is his actual recommendation for the weakness he diagnosed.

A1.5 What he says *not* to worry about ⚑

> *"Don't worry about all this details. No need to memorize it."* (microarray platform mechanics)

> *"Don't bother with the details, I don't want to scare you, but you just need to understand that LC is to separate the protein again so that the separated sample can be inject to the mass spec."*

> *"I admitted that I'm not the guy working on the next-generation sequencing, I'm just the user. I'm not the engineer that making the next-generation sequencing machine."*

These are explicit exam-scope statements. They are as informative as the positive signals: the lecturer is telling you that the *principle* is examinable and the *instrument minutiae* are not.

A1.6 His standing principle

> *"I always emphasize, seeing is believing."* — and then he goes and finds a paper to prove the point rather than asserting it. When he makes a claim in the exam, expect the claim to be defensible with an example you can produce.

A2 The three traps he named, in exam-ready form

Trap 1 — Positive vs negative CRISPR screen ⚑ (the named one)

This is the one he flagged. Both screens introduce the same two things to the cells: the **Cas9 enzyme** and the **sgRNA library**. The difference is what happens to the knockout cells after the drug treatment.

	Positive screen	Negative screen
What you are looking for	cells that **survive** the drug	cells that **die** because of the drug
What happens to the knockout	knockout **enriched** in the surviving population	knockout **depleted** — you cannot find it any more
Gene identified	gene whose loss makes cells **drug-sensitive** (gene A)	gene whose loss makes cells **drug-resistant** (gene B)
Reading it out	NGS shows gene A sgRNAs over-represented	NGS shows gene B sgRNAs under-represented

His own words for the negative case: *"once the gene B is knockout, the cell become fairly sensitive to the drug treatment and they all die. So since all the cell die, you cannot detect in here… So therefore, the gene B is identified to be responsible for the drug resistance."*

And for the positive case: *"if the cell with the gene A knockout, they can grow very well after the gene treatment. That means the gene A is responsible for the cell for this drug sensitivity."*

The third arm — the control. Without drug, gene C knockout cells are also gone. Gene C is simply essential for cell survival, and it is nothing to do with the drug. *"This control arm is very important, because without the control arms, you will mix up the finding with the cell survival and also the genes responsible for the drug resistance."*

The synonyms the literature uses. These screens are also called **positive selection / FOF** (fitness-of-fitness: the knockout is *enriched*) and **negative selection / dropout** (the knockout is *depleted*). His naming follows this standard convention, so you can use either vocabulary — but say which one you are using. The **safest sentence to write**, because it depends only on the readout and not on how you label the gene's role:

> "The screen identifies genes whose **loss changes fitness under the drug treatment**. A knockout that is **enriched** among survivors contributes to drug sensitivity (FOF / positive selection); a knockout that is **depleted** because the cells die contributes to drug resistance (dropout / negative selection). The untreated control arm separates genes that are simply essential for survival from genes that modify the drug response."

[MARK-WINNING LINE] "In a positive screen the knockout is enriched among survivors, so the gene confers drug sensitivity; in a negative screen the knockout is depleted because the knockout cells die, so the gene confers drug resistance. The untreated control arm separates genes that are simply essential for survival from genes that modify the drug response."

Trap 2 — NHEJ, not MMR (and say both limbs) ⚑

Defective **non-homologous end joining** → no V(D)J recombination → no TCR repertoire and no antibody repertoire → **severe combined immunodeficiency**.

Say both limbs. A complete answer names the **T cell receptor** *and* the **antibody from the B cell or plasma cell**. Do not write "NHEJ is defective, therefore no T cells"; write the full chain.

Note the link forward: NHEJ is the *same* pathway that CRISPR borrows. Cas9 makes a double-strand break; NHEJ repairs it error-precisely enough to create indels; the frame shift destroys the gene. A question can connect L4 to L6 — and this lecture does exactly that.

Trap 3 — Describe vs explain ⚑

The exam verb decides the answer shape. When the verb is **explain**, naming a mechanism is mandatory.

[DESCRIBE VS EXPLAIN, SIDE BY SIDE] *Describe (partial marks):* "The survival curve of the treated group drops more steeply and earlier than the control group." *Explain (full marks):* "The treated group drops earlier and more steeply, which indicates the gene is involved in the early steps of tumour progression rather than in late progression; this is consistent with a pro-tumour role of the target in this tumour type."

Same data. The second version earns the mark because it says *why the observation looks like that*.

Figure L3 — Positive and negative CRISPR screens: the same two inputs, opposite readouts

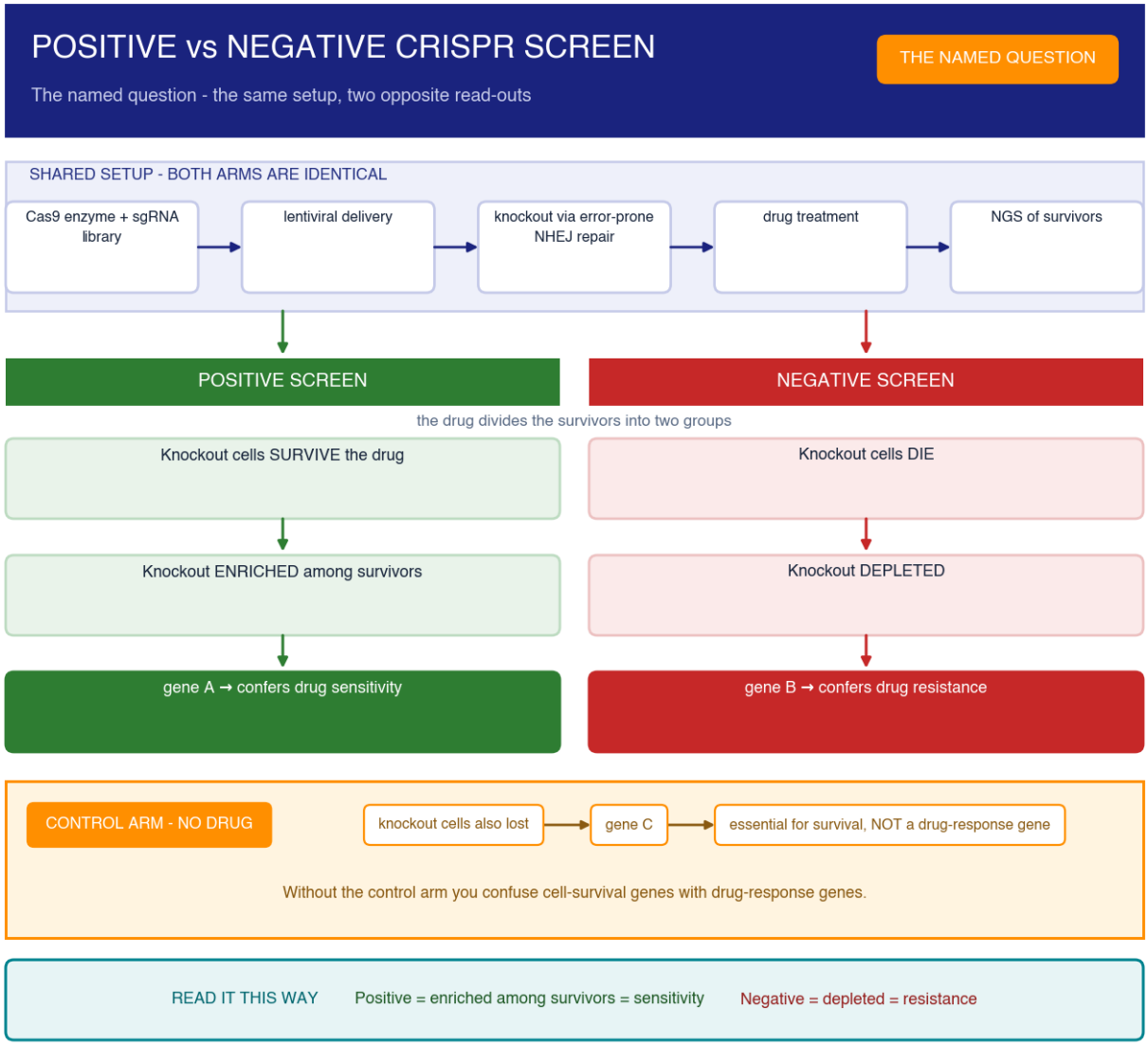


Figure L3. The positive screen (knockout enriched among survivors, so the gene confers drug sensitivity) against the negative screen (knockout depleted because the cells die, so the gene confers drug resistance), with the untreated control arm that separates a drug-response gene from an essential gene. Figure authored at 6.5 in print width and placed 1:1 on this page.

How to read this plate: read the control arm in the middle first - it is what stops you mixing up survival with drug response. Then read the positive and negative rows as mirror images. If you can state the readout and the gene's consequence in one English sentence, you can answer the named exam question.

A3 The numbers and thresholds you must be able to recall

Recalled numbers are cheap marks — but only the ones he actually put on a slide or spoke aloud. Here they are, with the context attached, because a number without its referent is worth nothing.

A3.1 Midterm and course logistics

Value	What it refers to
18.5	Median score on the midterm
29	Highest score on the midterm
1	The number of students who got the MCQ section fully correct
3 short questions	Midterm structure: Q1 NHEJ/SCID, Q2 done well, Q3 survival-difference explanation
2 classes	This lecture plus one more, on molecular biology techniques and their clinical applications, before returning to genetics
Phase 1b	His team's trial: Hong Kong, Shanghai, Australia
4 cities	Guangzhou, Chengdu, Shanghai, Beijing — PBMC collection for the HBV biomarker
2 central labs	Singapore (HK + Australia samples) and Shanghai (Chinese samples)

A3.2 Nucleic acid quality

Value	What it refers to
28S and 18S	The two dominant rRNA peaks that make a human RNA sample look intact on a Bioanalyzer
> 95 %	The fraction of total cellular RNA that the slide attributes to ribosomal RNA, and which must therefore be depleted before library prep. (The figure most often quoted for rRNA as a share of total RNA *mass* is ~80–85 %.)
RIN < 6 = poor quality	His slide's own cut-off, shown with three electropherograms: **intact RNA = RIN 10**, **partially degraded = RIN 5**, **strongly degraded = RIN 3**. Agilent's official scale is **1–10 with 10 = cleanest** (app note 5989-1165EN), which **agrees** with the slide. For FFPE, RIN is less informative because the ribosomal peaks are lost — **DV200** (>200 nt fraction) is the recommended FFPE metric.
28S : 18S ≈ 2 : 1	The expected ratio in intact mammalian RNA. A 1 : 1 ratio or a fainter 28S band indicates degradation.
Poly-A	The tail used to select mRNA; or to *remove* mRNA when you want the non-coding RNAs

A3.3 PCR

Value	What it refers to
Ct ~13–17	The Ct band quoted on his slide for house-keeping genes used to normalise real-time PCR (the slide writes it with a tilde — it is approximate). ⚠ Strictly this describes the **most abundant** reference genes (β-actin ≈ 13.9–18.6); across a six-reference-gene panel the usable band is roughly **13.9–26.1**.
2 components	What makes TaqMan specific: a **primer** *and* a **probe**; the probe carries a reporter **Q** (quencher) until the target is amplified
4 components	The PCR reaction: DNA template, primers, dNTPs, DNA polymerase

[DIRECTION TO REMEMBER] the **larger the Ct value, the LOWER the copy number** of the target in the original sample — more cycles were needed to reach threshold. Small Ct = abundant target.

A3.4 Genomics and transcriptomics

Value	What it refers to
100 / 1,000 / 10,000	How many genes a microarray measures simultaneously
21 cancer types, 316 patients, 397,810 T cells	The pan-cancer single-cell landscape of tumour-infiltrating T cells (Zheng L et al., 2021)
16 h	The media-control incubation in the multi-tissue T-cell study
5 min / 1 week	A break, and the time since the midterm was held (not examinable content — listed so you do not mistake it for data)

The microarray vs RNA-Seq comparison — the numbers on his slide.

He framed this comparison as the justification for abandoning microarrays, so understand the *direction* of every column rather than memorising every cell:

[CITATION NOTE] The slide credits *"Rao MS et al., 2019"* — it does **not** name a journal. The source is **Rao MS, Van Vleet TR, Ciurlionis R, Buck WR, Mittelstadt SW, Blomme EAG, Liguori MJ (2019), *Comparison of RNA-Seq and Microarray Gene Expression Platforms for the Toxicogenomic Evaluation of Liver From Short-Term Rat Toxicity Studies*, *Frontiers in Genetics* 9:636, doi:10.3389/fgene.2018.00636** (PMID 30723492). **His slide reproduces the paper's Table 1A and Table 1C exactly**, so the numbers on the slide *are* the published numbers. The one place the slide stops short is the scatter: it plots **four** treatments, so DCLF appears only in the tables — the paper reports **Spearman's ρ = 0.30** for DCLF (and 0.44 for APAP, i.e. the paper uses Spearman's ρ for those two). **In the exam, quote his slide.** If you are asked to cite the source, cite *Frontiers in Genetics* 2019. See Appendix C.

Treatment	Microarray DEGs (total)	RNA-Seq DEGs (total)	R on his slide
ANIT	1014	3139	0.65
APAP	92	512	0.44
CCl_4	689	2127	0.63
DCLF	16	196	— (not plotted)
MDA	1081	3672	0.67

The slide also gives the up/down split: microarray ANIT 507 + 507, APAP 54 + 38, CCl_4 361 + 328, DCLF 3 + 13, MDA 569 + 512; RNA-Seq ANIT 1634 + 1505, APAP 275 + 237, CCl_4 1091 + 1036, DCLF 93 + 103, MDA 1899 + 1773. The five treatments are rat-liver hepatotoxicants, and APAP and DCLF produce far fewer differentially expressed genes than the other three — which is why the paper's histopathology for those two is the mildest.

And the dynamic range table (same treatments): microarray max fold-change was far smaller than RNA-Seq's — for example ANIT **41** (array) vs **577.5** (RNA-Seq), and MDA **15.7** vs **226.5**.

[THE POINT, IN ONE SENTENCE] the platforms agree on what is expressed (R ≈ 0.44–0.67), but RNA-Seq detects **more** differentially expressed genes and with **much larger** fold-changes, because the microarray saturates on abundant transcripts and loses signal on rare ones.

A3.5 Proteomics

Value	What it refers to
5–250 kDa	The size range covered by SDS-PAGE
~25 kDa / ~50 kDa	IgG **light chain** / **heavy chain** on a reducing gel
~150 kDa	The intact IgG molecule on a non-reducing gel
2 dimensions	2D-PAGE: first dimension = **pI**, second dimension = **molecular weight**
2 amino acids	The fragment size that MS/MS resolves, enabling de novo sequence assembly
2 components	Biomarker discovery = **protein separation** (2D-PAGE) + **protein identification** (mass spec)

A3.6 Immunoassays, sequencing and protein expression

Value	What it refers to
10–20 proteins	How many proteins a multiplex assay measures in a single sample
20–30 peptides	Candidate neoantigen peptides predicted, then screened in vitro before scale-up
~10 years	How long ago it was nearly impossible to deduce both TCR and neoantigen sequences
3 functions	Why you add an Fc fusion tag: **stability**, **detection**, **dimer formation**
2 methods	Getting DNA into a mammalian cell: **transfection** and **electroporation**
2 plates	The two metal plates across which voltage is applied in an electroporation cuvette
Gly_3Ser	The flexible linker between a fusion tag and your protein

Which of the three Fc functions is the most important? He was explicit: *"The third one is more important because it can affect the function of your target protein."* Nuclear receptors and HER2 function as **dimers**; the monomeric form is non-functional. The Fc region carries an inter-chain **disulfide bond**, so fusing your target to Fc forces a **homodimer**. If a question asks which fusion strategy suits a dimerising target, the answer is **Fc**, and the reason is *functional dimer formation* — not convenience.

Figure L4 — Microarray against RNA-Seq: what each platform can fail to do

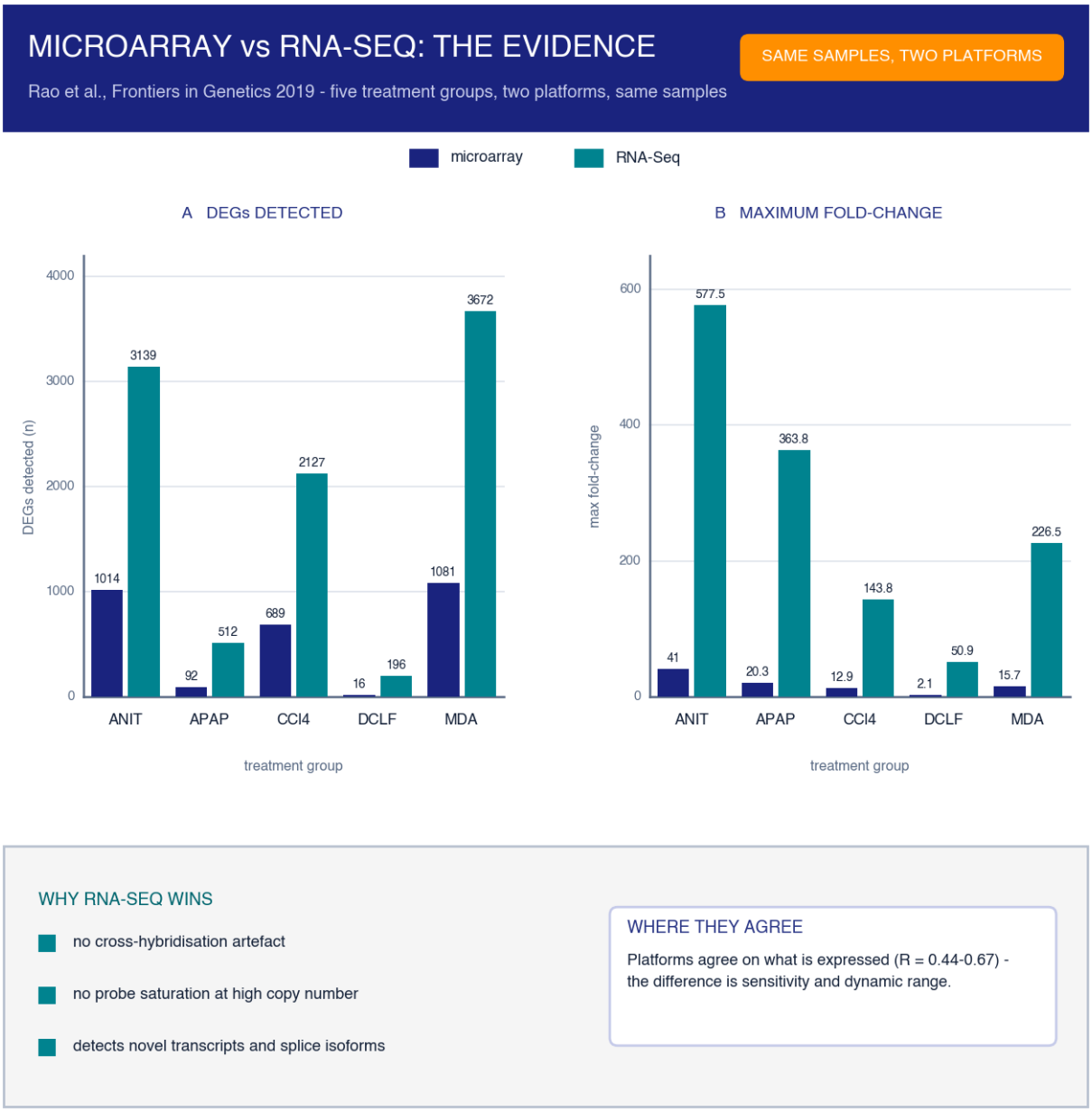


Figure L4. The comparisons the lecturer made: what is on the chip, saturation on abundant transcripts, cross-hybridisation, and novel transcripts and splice isoforms - with the correlation and fold-change evidence behind them. Figure authored at 6.5 in print width and placed 1:1 on this page.

How to read this plate: every row is one 'what it cannot do' statement. Learn the limitation on the left and the reason on the right. The numbers underneath are the evidence for the direction, not the point - the direction is what earns the mark.

A4 The discriminator pairs you must be able to separate

The exam will not ask you to define a technique. It will ask you to choose between two, and the mark is for the **reason**. Each row below is one line you must be able to say out loud without hesitating.

No.	Pair	The discriminator you must be able to state
1	**FFPE vs fresh frozen**	FFPE preserves architecture for histology/IHC but degrades nucleic acid; fresh frozen is the right material for genomic and transcriptomic profiling.
2	**Serum vs plasma**	Serum is clot-derived (no anticoagulant); plasma is anticoagulated. Analyte levels can differ significantly between them, so the matrix must be stated and chosen deliberately.
3	**SYBR Green vs TaqMan**	SYBR Green is cheap and sequence-agnostic (binds any dsDNA) so it loses specificity to non-specific priming; TaqMan adds a target-specific **probe** with a reporter and a **quencher**, so signal only appears when the target is amplified.
4	**Multiplex vs single-target PCR**	Multiplex detects several targets in one reaction — chosen to **minimise variation between tests** and to **save scarce sample**, not to save money (it is usually more expensive).
5	**Microarray vs RNA-Seq**	Microarray measures known sequences by hybridisation and saturates on abundant transcripts; RNA-Seq sequences everything, has no cross-hybridisation artefact, quantifies low and high expression, and finds novel transcripts and splice isoforms.
6	**Bulk RNA-Seq vs scRNA-Seq**	Bulk averages every cell in the sample so one cell type's signal is masked; scRNA-Seq resolves the transcriptome of individual cells, which is what you need for heterogeneity.
7	**ChIP-Seq vs RNA-Seq**	ChIP-Seq asks *where* a protein binds the DNA (using an antibody to pull down the DNA–protein complex); RNA-Seq asks *how much* RNA is present.
8	**1D SDS-PAGE vs 2D-PAGE**	1D separates by molecular weight only; 2D separates by pI first and molecular weight second, giving far higher resolving power.
9	**Reducing vs non-reducing SDS-PAGE**	Reducing (DTT) breaks the disulfide bridges so IgG separates into 50 kDa heavy and 25 kDa light chains; non-reducing keeps the intact ~150 kDa molecule together.
10	**IEF vs SDS-PAGE**	IEF separates by isoelectric point — the pH at which net surface charge is zero, so the protein stops migrating; SDS-PAGE separates by size once the protein has been denatured to a linear form.
11	**LC-MS vs LC-MS/MS**	LC-MS gives a peptide fingerprint that must match a database, so it **cannot identify a novel protein**; LC-MS/MS fragments to two amino acids and performs **de novo sequencing**, so it can.
12	**Direct vs indirect ELISA**	Direct has an enzyme-conjugated detection antibody (one antibody, one signal); indirect uses an unconjugated primary plus enzyme-conjugated secondaries, so **several secondaries bind one primary — signal amplification**, which is what you want for low-abundance antigen.
13	**ELISA vs ELISpot**	ELISA measures the **amount of soluble protein** in a liquid sample; ELISpot measures the **number of antigen-specific cells** by capturing the cytokine (IFN-γ) each cell secretes.
14	**Positive vs negative CRISPR screen**	Positive = knockout **enriched** among survivors → gene confers drug **sensitivity**. Negative = knockout **depleted**, cells die → gene confers drug **resistance**. Always state the untreated control arm.
15	**Transfection vs electroporation**	Transfection packages DNA in lipid vesicles (DNA is charged and cannot cross the hydrophobic bilayer); electroporation makes transient pores with a voltage pulse. Lipid transfection fails on **T cells, NK cells and macrophages** — use electroporation there.
16	**Fc fusion vs GFP tag**	Fc is for **stability, detection and forced dimerisation** (functional); GFP is for **localisation and co-localisation** (observational), and can also improve solubility and stability.
17	**Personalised vs off-the-shelf cancer vaccine**	Personalised is built from one patient's own mutations; off-the-shelf uses **shared epitopes** found by sequencing a large cohort of the same cancer type.

[HOW TO USE THIS TABLE] Do not read it as a list to memorise. For each row, say the discriminator aloud as one sentence in English. If you cannot, that row is not yet learned.

A5 Method selection: "given this question, which technique?"

This is the shape a scenario question takes. The marks are for the *choice* **and** the *reason* **and** the *limit*. Study the reason column; the choice follows from it.

If the question is…	Use	Because	And remember it cannot…
Is my sample good enough to profile?	**Bioanalyzer RIN / 28S-18S trace**	Two sharp rRNA peaks mean intact RNA; FFPE gives a smear	tell you whether the analyte is *clinically* meaningful
Which gene is expressed, and how much?	**RNA-Seq**	sequences everything; no saturation, no cross-hybridisation	detect a very rare transcript as reliably as an array can

If the question is…	Use	Because	And remember it cannot…
Which genes are expressed, known set only?	**Microarray**	cheap, fast, still used clinically (MammaPrint)	find a novel transcript or a novel splice isoform
Which cell type is doing it?	**scRNA-Seq**	one cell, one transcriptome; resolves heterogeneity	be cheap — cost and bioinformatics load are real
Where does this transcription factor bind?	**ChIP-Seq**	antibody pulls down the DNA–protein complex, then sequence it	stand alone — **you must validate with a wet-lab reporter assay**
How much of this mRNA is there?	**real-time PCR**	Ct converts to relative or absolute quantity	tell you the *isoform* unless you design for it
Is the signal specific?	**TaqMan probe**	reporter is quenched until the target is cleaved	be cheap or quick to design
Several targets, one scarce sample?	**multiplex PCR**	minimises between-test variation and saves sample	be cheaper than single-target
What is the mass of this protein?	**SDS-PAGE**	denatured to linear form, migration depends on size alone	separate isoforms of identical mass
What is this protein's pI?	**isoelectric focusing**	protein migrates until net charge is zero	resolve proteins of the same pI
Both pI and mass, high resolution?	**2D-PAGE**	two orthogonal separations	identify the protein by itself — you still need mass spec
Which protein is in this spot?	**LC-MS**	peptide fingerprint matches a database	identify a **novel** protein
Which novel protein is in this spot?	**LC-MS/MS**	de novo sequencing from two-amino-acid fragments	work without any effort at de novo assembly
How much soluble protein?	**ELISA**	quantitative, works on serum/plasma	count cells
How many antigen-specific T cells?	**ELISpot**	counts IFN-γ-secreting cells	give a concentration
Which gene gives drug resistance?	**CRISPR negative screen**	knockout cells die, so they are depleted	tell you the effect without a control arm
Which gene gives drug sensitivity?	**CRISPR positive screen**	knockout cells survive, so they are enriched	distinguish a drug-response gene from an essential gene without a control
How do I get DNA into a T cell?	**electroporation**	lipid transfection does not work on primary T/NK/macrophages	be as gentle as transfection
Will my protein dimerise?	**Fc fusion**	Fc forces a disulfide-linked homodimer	leave your protein's function untouched
Where is my protein?	**GFP / RFP tag**	fluorescence reports localisation and co-localisation	tell you whether the protein is *functional*
Which antigen does this tumour present?	**TCR-seq + neoantigen prediction**, or **immunopeptidomics (MS)**	one infers the antigen from the TCR, the other reads it directly	skip the in vitro validation — you must still screen the peptides

Figure L2 — Method selection: given the question, which technique?

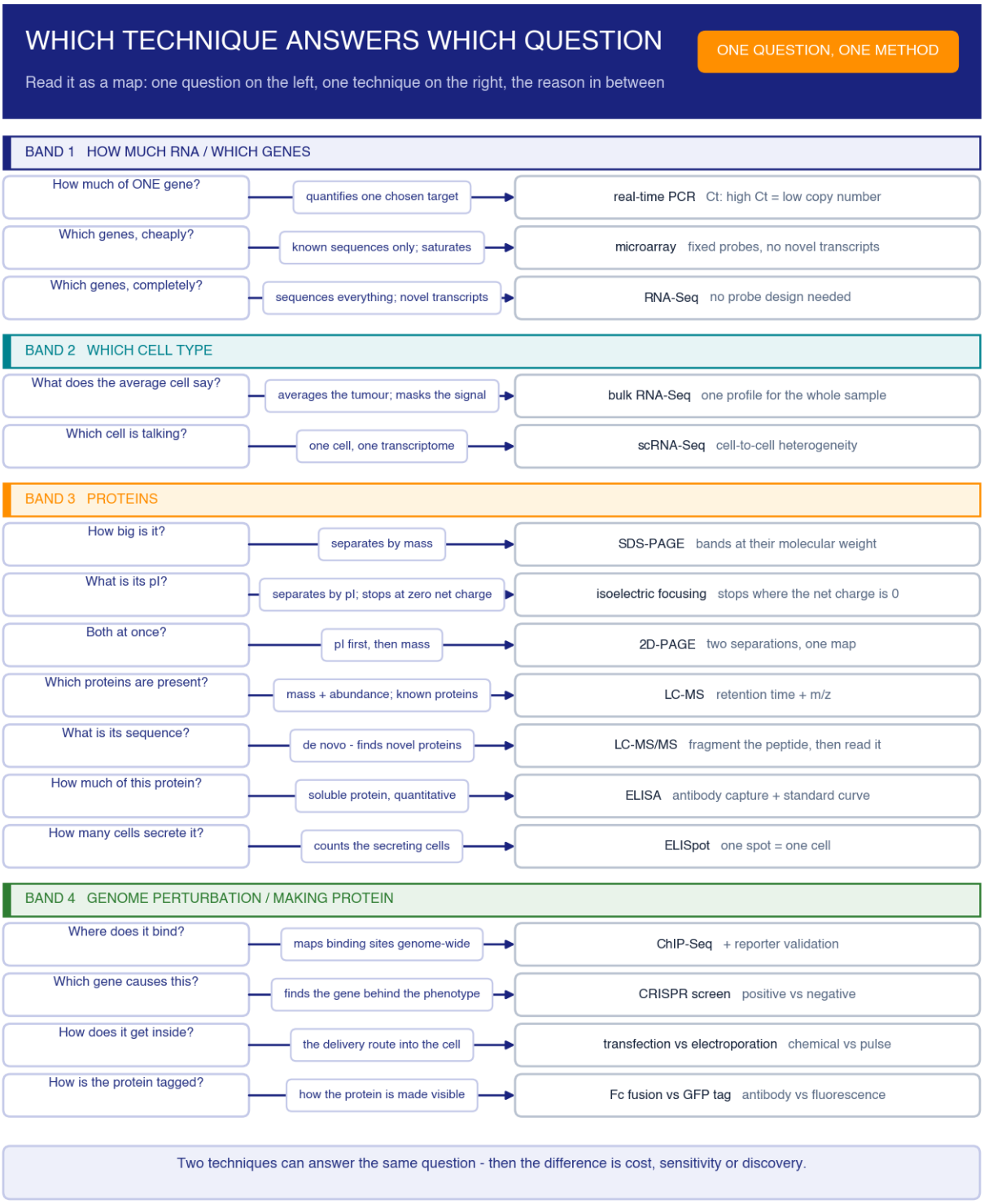


Figure L2. The scenario-question map: the question you are asked, the technique that answers it, the reason, and the one thing that technique still cannot do. Figure authored at 6.5 in print width and placed 1:1 on this page.

How to read this plate: cover the right-hand column and work left to right - question, technique, reason. Then uncover it and check the limit. The marks are for the choice and the reason; naming the limit is what separates a strong answer from a complete one.

A6 The English answer skeleton, and the habits that earn marks

Dr. Wong told you the marks are not for description and not for vocabulary. Here is the shape he was describing.

A6.1 The four moves

Every short answer, in this order:

1	**State the observation.** One sentence, with the number in it if there is a number. *"The treated group shows a steeper and earlier fall in survival than the control group."*
2	**Name the mechanism.** This is the part students omit, and it is where the marks are. *"This indicates that the target is engaged in the early steps of tumour progression."*
3	**Interpret the clinical meaning.** What it means for the patient or the assay. *"The target is therefore a candidate for early-stage intervention rather than a marker of late disease."*
4	**State the action or the limit.** What you would do, or what you cannot conclude. *"This conclusion is specific to this tumour type and cannot be generalised across tumour types."*

[THIS IS THE FIX FOR THE PROBLEM HE DESCRIBED] A student who writes move 1 and stops has described. A student who writes moves 1–2 has explained. Moves 3 and 4 are what separate a good answer from a complete one.

A6.2 Marker-pleasing habits

- **State the number.** If the question supplies a number, use it. A number in your answer shows you read the graph.
- **Name both the mechanism and the molecule.** "The target" is worth nothing. "The target is a receptor tyrosine kinase that is amplified at the 17q12 locus" is worth marks.
- **Say what the method cannot do.** Every technique has one limit, and naming it proves you know the method rather than the slogan.
- **Write complete sentences.** No bullet fragments, no arrows, no "NHEJ → SCID" on its own.
- **Answer in English.** Always. Zero marks otherwise.

A6.3 Two templates

Template A — a technique-comparison question.

"Both [X] and [Y] can answer [the question]. [X] works by [mechanism], so it is [advantage]; however it cannot [limit]. [Y] works by [mechanism], so it is [advantage]; however it cannot [limit]. For the purpose of [the stated goal], I would choose [Y], because [reason tied to the goal]."

Template B — a "which gene explains this phenotype" question.

"The [positive/negative] screen identifies a gene whose [knockout is enriched among survivors / knockout is depleted because the cells die], so gene [X] [confers drug sensitivity / confers drug resistance]. The untreated control arm shows that gene [C] knockouts are lost even without drug, so gene [C] is essential for survival and is not a drug-response gene. The next step is target validation in an orthogonal assay, followed by pathway enrichment analysis of the genes that co-vary with [X]."

A7 What NOT to memorise

He explicitly released these. Studying them is wasted time — and, worse, it is the difference between knowing the principle and not knowing it.

Do not memorise	Why
Microarray platform mechanics (two-colour vs one-colour workflow detail, scanning hardware)	*"Don't worry about all this details. No need to memorize it."*

Do not memorise	Why
Chromatographic instrumentation in LC	*"Don't bother with the details, I don't want to scare you."*
How NGS machines are built	*"I'm not the guy working on the next-generation sequencing, I'm just the user."*
Every cell of the microarray vs RNA-Seq number tables	Know the **direction** and the order of magnitude. The tables are evidence for a point, not the point.
The clinical-trial logistics market figures on the slide	Commercial background, not molecular biology.
Exact instrument settings, buffer compositions, run conditions	Not examinable; they change every year.

[THE CAVEAT, IN HIS SPIRIT] *"don't memorise the glycosylase chemistry"* does **not** mean *"don't know what BER repairs."* The same applies here — do not memorise the platform, but you must know **what each technique measures and what it therefore cannot tell you**.

A8 Self-test — 15 questions, closed book, English

Answer all fifteen in English, in writing, without looking. Then mark yourself with the key.

Q1	You receive a tumour block from a pathologist and must run expression profiling. What do you ask for, and what does the Bioanalyzer trace of a good sample look like?
Q2	A biomarker for rheumatoid arthritis is undetectable in plasma but significantly elevated in serum. Explain why, and say what you must state in an exam answer that mentions a blood biomarker without naming the matrix.
Q3	You want to know where the transcription factor TEAD binds the genome. Name the technique, the three steps, and the one experiment you must do afterwards to validate the result.
Q4	State the difference between the direct and the indirect ELISA, and say which you would use for a low-abundance circulating biomarker and why.
Q5	In a pooled CRISPR knockout screen, the sgRNAs targeting gene B are almost absent after drug treatment, but they are abundant in the untreated control. Is gene B a drug-sensitivity gene or a drug-resistance gene? Name the screen type, and say what the control arm is for.
Q6	In a separate screen, gene A sgRNAs are enriched after drug treatment. What does gene A confer, and what is this screen called?
Q7	A student writes "NHEJ deficiency causes SCID because the cells cannot make B cells." Why does this lose marks, and what is the complete version?
Q8	Explain the difference between microarray and RNA-Seq in terms of *what each can fail to do*.
Q9	Explain, with a mechanism, why a microarray gives a poor measurement for a very high-abundance transcript and for a very low-abundance transcript.
Q10	Why can LC-MS not identify a novel protein, and what does LC-MS/MS do differently?
Q11	On a reducing SDS-PAGE gel, an antibody sample gives two bands. State their molecular weights, the chemical that produced them, and what the same sample looks like on a non-reducing gel.
Q12	In isoelectric focusing, in which direction does a protein migrate, and what stops it?
Q13	You need to engineer a primary human T cell with an expression plasmid. Why not use lipofectamine? What do you use instead, and what is the physical mechanism?
Q14	You are expressing a protein that is only functional as a dimer, and you want to detect whether it was expressed at all. Which fusion tag, and what are its three functions in order of importance?
Q15	Write the four moves of the English answer skeleton, and the sentence frame you would use to finish a short answer about a survival curve.

Answer key

1	Ask for **fresh frozen** tissue — FFPE crosslinks and degrades nucleic acid, so profiling data from FFPE is poor. A good RNA sample shows **two sharp peaks at 28S and 18S** on the Bioanalyzer; degraded FFPE RNA gives a smear with no clean peak.
2	Serum is derived from **clotting** (no anticoagulant) and plasma from **anticoagulated** blood; the analyte's stability and half-life differ between the two matrices, so the same target can behave completely differently. In an answer, **state the matrix explicitly** — and if the question omits it, say that the result cannot be interpreted without knowing it.

3	**ChIP-Seq.** Three steps: (i) **isolate** the specific DNA–protein complex with an antibody against the factor (TEAD), (ii) **sequence** the recovered DNA, (iii) **bioinformatics** to map the genomic locations (promoter, intronic, intergenic) of the target genes. Validation: clone the candidate sequence upstream of a **luciferase reporter**, introduce it into a cell, add or express TEAD, and show the reporter is activated.
4	Direct: the detection antibody itself is **enzyme-conjugated** (e.g. HRP) — one antibody, one signal. Indirect: the detection antibody is **not** conjugated; a second, enzyme-conjugated antibody binds it, and **several secondaries can bind one primary**, so the signal is amplified. Use **indirect** for a low-abundance circulating biomarker, because amplification raises the signal-to-noise ratio at low concentration.
5	Gene B is a **drug-resistance** gene: knocking it out makes the cells drug-sensitive, so they die and the sgRNAs are **depleted**. This is a **negative screen**. The control arm (no drug) shows whether a knockout is lost because of the drug or simply because the gene is essential for survival.
6	Gene A sgRNAs are **enriched**, meaning cells with gene A knocked out **survive** the drug — so gene A confers **drug sensitivity**. This is a **positive screen**.
7	It names only one limb of the answer. The complete version: defective **NHEJ** prevents **V(D)J recombination**, so there is **no T cell receptor repertoire and no antibody repertoire** (the antibody comes from the **B cell / plasma cell**) — hence **severe combined immunodeficiency**. (MMR is not the answer.)
8	Microarray measures only **known sequences by hybridisation**, so it can miss novel transcripts and novel splice isoforms, and it **saturates** on abundant transcripts so it cannot quantify very high copy number correctly. RNA-Seq **sequences everything**, has no cross-hybridisation artefact, quantifies low and high expression well, and detects novel transcripts and alternative splice isoforms.
9	Microarray signal depends on **hybridisation of a labelled probe to a fixed, saturable amount of target on the array**. For a **very abundant** transcript all probes are saturated, so extra copies produce no extra signal (the curve is flat — under-estimated). For a **very rare** transcript hybridisation is inefficient, so the signal is under-estimated and the gene may be missed altogether. RNA-Seq avoids both by counting reads.
10	**LC-MS** produces a peptide **mass fingerprint** which must be **matched against a database**; a novel protein, or a novel splice variant, is by definition absent from the database, so it cannot be identified. **LC-MS/MS** takes the peptides selected in the first MS and fragments them in a second MS down to two-amino-acid fragments, from which the sequence is assembled **de novo** — no reference database required.
11	**~25 kDa (light chain)** and **~50 kDa (heavy chain)**, produced by adding a **reducing agent (e.g. DTT)** that breaks the inter-chain **disulfide bonds**. Non-reducing, the chains stay joined and the intact IgG appears as a single band at **~150 kDa**.
12	It migrates through the pH gradient and **stops at its isoelectric point**, the pH at which its net surface charge is zero (at high pH the protein is negatively charged and moves toward the anode; at low pH it is positively charged).
13	Lipid transfection is inefficient on primary immune cells — **T cells, NK cells and macrophages** — because they are difficult to transfect. Use **electroporation**: a rapid voltage pulse across the membrane creates **transient pores**, DNA enters during the pore opening, and the membrane reseals when the pulse ends.
14	**Fc fusion protein.** Functions in order of importance: (1) **forced dimer formation** — the Fc region forms an inter-chain disulfide bond, so your target is produced as a homodimer, which is required for nuclear receptors and HER2; (2) **stability**; (3) **detection** — if you have no antibody to your own target, use an anti-Fc antibody to show it was expressed. (GFP is the alternative when the question is *localisation*.)
15	**1** State the observation (with the number). **2** Name the mechanism. **3** Interpret the clinical meaning. **4** State the action or the limit. Finishing frame: *"The curve separates early and steeply in the treated group, indicating that the target is involved in the early stages of progression; this is specific to this tumour type and should not be generalised across tumour types."*

PART B — LEARN FROM ZERO

B0 What you need before you start

If any of the following feels shaky, fix it first. Each one is a Lecture 1–5 concept, and each one is used in Lecture 6.

Prerequisite	Where you met it	Why Lecture 6 needs it
DNA structure, base pairing, replication	L3, L4	You must understand primers, templates and the polymerase
PCR: template + primer + dNTP + polymerase	L4	Real-time PCR is the whole of block 2
NHEJ and the error-prone repair of a double-strand break	L4	CRISPR works *because* NHEJ repairs Cas9's cut badly — and L4's SCID question is on the L6 exam
Central dogma, transcription, RNA processing	L1, L5	cDNA, mRNA, poly-A, RNA-seq all assume it
Antibody structure: heavy chain, light chain, disulfide bonds	L2, L4	The reducing vs non-reducing gel is an antibody question
T cell biology: TCR, CD3, CD28, MHC, antigen presentation	L2, L4	Clonal expansion, cancer vaccines, ELISpot
Cell cycle and mitosis	L2	The contrast between bulk and single-cell resolution
Basic chemistry: pH, charge, molecular weight	L1	Isoelectric focusing is a pH question

If you are shaky on **NHEJ** or on **T cell receptor/antibody structure**, start there — those two produce marks on both Lecture 4-style questions and Lecture 6 questions.

B1 The ten concept chains

Read a chain in order. Each one ends with the single comparison you must be able to make out loud, because that is what the exam asks for.

Chain 1 — The sample gates everything

The one-line idea: You cannot extract information that was never preserved. Sample handling is not clerical; it is the first experimental variable.

The chain:

```
Tissue arrives
↓
Is it FFPE or fresh frozen? → FFPE: good for histology/IHC, poor for profiling → ask for FF
↓
Is it a homogeneous tissue? → a tumour is tumour + normal + immune → bulk profiling masks the signal
↓
Is the RNA intact? → Bioanalyzer: 28S + 18S sharp = good; smear = degraded
↓
Blood? which matrix? → serum (clotted) vs plasma (anticoagulated) → the RA example: opposite results
↓
Multi-site study? → dry run first (ship fakes, measure PBMC viability)
↓
```

```
One central lab → removes operator/centre/day variation
↓
Regulatory reality → samples often cannot leave the country → second central lab → cost
```

The comparison you must say out loud: *"FFPE tissue is the right material for histology because it preserves the architecture, but it is the wrong material for genomic profiling because the nucleic acid is degraded — the same principle as 'garbage in, garbage out'."*

Self-check. Why did the horizontal tube beat the vertical one in the dry run? → *More gas exchange with the blood and the cells do not settle and pack against the bottom, so PBMC viability is higher.*

Chain 2 — From tissue to a number: real-time PCR

The one-line idea: PCR turns a vanishingly small amount of RNA into something you can measure, and the measurement is a *threshold crossing*, not an absolute reading.

The chain:

```
RNA from the sample
↓
Reverse transcription → cDNA (this is why RNA quality matters so much: bad RNA → bad cDNA)
↓
PCR: template + forward/reverse primers + dNTPs + DNA polymerase + amplicon
↓
Fluorescence accumulates each cycle
↓
Ct = the cycle at which fluorescence crosses threshold
↓
Ct is INVERSELY related to starting copy number
high Ct → few target copies in the sample
low Ct → many target copies
↓
Normalise against house-keeping genes (Ct ~13–17)
↓
Two chemistries: SYBR Green (cheap, non-specific) vs TaqMan (probe + quencher, specific)
↓
Multiplex: several targets in one reaction
```

The comparison you must say out loud: *"SYBR Green fluoresces when it binds any double-stranded DNA, so a non-specifically amplified product also produces signal. TaqMan adds a sequence-specific probe whose reporter is held silent by a quencher until the target is amplified and the probe is cleaved, so signal means the intended product — at the cost of more design effort and expense."*

Self-check. Why can you not compare a Ct of 18 from one gene with a Ct of 18 from another gene? → *Ct is only meaningful relative to the same reference and the same assay; you compare ΔCt between samples, not raw Ct across targets.*

Chain 3 — Many genes at once: three different questions

The one-line idea: Microarray, RNA-Seq and scRNA-Seq look like a family, but they answer three different questions. Learn them as answers, not as a list.

The chain:

```
"What genes are involved?"
↓
Do you already know which sequences to look for?
├─ YES → MICROARRAY: probes on a chip, hybridisation, fluorescence
│ can find NOTHING you did not put on the chip
│ saturates on abundant transcripts; cross-hybridisation artefact
│ still clinically alive: MAMAprint in breast cancer
└─ NO → RNA-SEQ: sequence everything
no saturation, good at low and high expression
finds novel transcripts and alternative splice isoforms
needs bioinformatics; costlier

"Which CELL TYPE is doing it?"
↓
BULK RNA-Seq averages every cell → a minority cell type's signal is masked
↓
scRNA-Seq: isolate single cells → one transcriptome per cell
isolation methods: manual pipetting · laser microdissection · FACS · microfluidics
droplet: cells + barcoded beads + oil → lysis → RT → library → sequencing
```

The comparison you must say out loud: *"A microarray can only report transcripts you chose in advance, and it saturates on abundant transcripts, so it both under-detects rare transcripts and under-quantifies very abundant ones. RNA-Seq sequences everything, so it detects novel transcripts and splice isoforms and quantifies across the full dynamic range. And neither resolves which cell type is expressing — only single-cell sequencing does that."*

Self-check. The Rao 2019 comparison (*Frontiers in Genetics* 9:636): what did the correlation between platforms tell you, and what did the DEG counts tell you? → *The platforms agree on what is expressed ($R \approx 0.44$–0.67), but RNA-Seq found many more differentially expressed genes with far larger fold changes (e.g. ANIT: 1014 array DEGs vs 3139 RNA-Seq DEGs; MDA max fold change 15.7 vs 226.5).*

Chain 4 — Finding where a protein binds: ChIP-Seq

The one-line idea: ChIP-Seq turns "I think this factor binds here" into a map, and then you have to prove it.

The chain:

```
Question: where does TEAD bind?
↓
Enhancers and repressors can sit FAR from the gene they control
→ you cannot just guess from the promoter
↓
1 ISOLATE: antibody against TEAD → captures the DNA–TEAD complex
(capture via biotin–streptavidin, or magnetic beads)
```

```
↓
2 DIGEST: protease removes the protein → free DNA fragments
↓
3 SEQUENCE → bioinformatics: map to genome
(promoter? intronic? intergenic?) → target genes
↓
VALIDATE (mandatory): clone the candidate sequence upstream of a LUCIFERASE reporter
→ transfect into CHO cells → add or express TEAD
→ if the reporter lights up, the binding is real
```

The comparison you must say out loud: *"ChIP-Seq tells you where a protein binds the DNA; RNA-Seq tells you how much RNA is present. And the sequencing result is a hypothesis, not a conclusion — it must be validated experimentally with a reporter assay before you believe it."*

Self-check. Why does ChIP-Seq need an antibody, and what does that antibody actually pull down? → *The antibody binds the transcription factor (e.g. TEAD), which brings the DNA–protein complex down with it; the protein is then digested away so only the bound DNA fragments are sequenced.*

Chain 5 — From a T cell to a cancer vaccine

The one-line idea: This is the chain the lecturer says is "really encouraging" — sequencing turns an immune response into a drug target.

The chain:

```
Tumour (or tumour-infiltrating lymphocytes, TIL) + blood
↓
Isolate T cells → single-cell sequencing (library prep can be restricted to TCR genes only
— you do not need whole-genome sequencing)
↓
TCR sequence → which clones expanded?
(clones share a TCR because one activated T cell proliferates)
↓
Why does that matter? A T cell's TCR physically interacts with its antigen
↓
So: TCR sequence → COMPUTER PREDICTION of the antigen (neoantigen) sequence
↓
Two independent routes to the antigen:
ROUTE 1 TCR-seq → predict antigen
ROUTE 2 immunopeptidomics → pull down the MHC–peptide complex directly → digest → MS/MS → de novo
↓
20–30 candidate peptides predicted
↓
MUST validate in vitro: peptide + PBMC/T cells → IFN-γ ELISpot; flow cytometry for CD8 and markers
↓
Scale up → personalised vaccine (one patient) OR off-the-shelf (shared epitope across a cohort)
```

The comparison you must say out loud: *"A personalised vaccine is built from one patient's own mutations; an off-the-shelf vaccine uses epitopes shared across a large cohort of the same cancer type. Both require in vitro validation of the predicted peptides before scale-up — prediction alone proves nothing."*

Self-check. Why does anti-CD3 plus anti-CD28 treatment appear in the workflow? → *CD3 engagement delivers the T cell receptor signal (signal 1) and CD28 delivers the co-stimulatory signal (signal 2); together they maintain survival and activate the cells so their transcriptome can be studied.*

Chain 6 — Separating proteins

The one-line idea: Every protein separation step removes one variable so the next one can be meaningful.

The chain:

```
Problem: proteins fold into different shapes, so migration would depend on shape too
↓
SDS-PAGE: denature to a LINEAR form with detergent (+ reduce disulfide bonds with DTT)
→ now migration depends ONLY on molecular weight
→ big protein = short distance; small protein = long distance (band near the bottom)
→ IgG example: reducing → 25 kDa light + 50 kDa heavy; non-reducing → 150 kDa intact
↓
But that only separates by size. What if two proteins have the same size?
↓
IEF: separate by CHARGE
protein charge depends on pH: high pH → negative; low pH → positive
pI = the pH where net charge is ZERO → the protein stops moving
↓
2D-PAGE: 1st dimension = pI, 2nd dimension = molecular weight
a spot now means (a specific pI, a specific mass)
↓
2D-DIGE: label sample 1 with Cy3 (green), sample 2 with Cy5 (red), run them on ONE gel
green only = higher in sample 1 · red only = higher in sample 2 · yellow = equal
this removes inter-gel variation, exactly like the one-colour vs two-colour array choice
```

The comparison you must say out loud: *"SDS-PAGE separates by size once the protein is denatured to a linear form; isoelectric focusing separates by charge and stops each protein at its isoelectric point, where its net charge is zero. 2D-PAGE couples them — first pI, then mass — which is why it has far higher resolving power, and why 2D-DIGE labels two samples with Cy3 and Cy5 onto a single gel to avoid inter-gel variation."*

Self-check. Why must the protein be denatured before measuring its size by electrophoresis? → *Otherwise migration would depend on the native shape and quaternary structure as well as mass, so the measurement would not be interpretable.*

Chain 7 — Identifying proteins: mass spectrometry

The one-line idea: Separation tells you *that* there is a protein; mass spectrometry tells you *which* — and only the tandem version can do it for something new.

The chain:

```
A 2D gel spot that differs between normal and tumour
```

```
↓
Cut out the spot → digest with TRYPSIN → peptide fragments
↓
LC (liquid chromatography): separate the peptides by hydrophobicity/solubility
↓
Route A — LC–MS: generate a peptide MASS FINGERPRINT → search a public database
works for known proteins
FAILS for a novel protein or a novel splice variant — nothing to match
↓
Route B — LC–MS/MS: first MS selects the peptide, SECOND MS fragments it
fragments resolve down to TWO AMINO ACIDS
computer assembles the sequence from the fragment series
= DE NOVO PEPTIDE SEQUENCING — no database needed
this is how you find a neoantigen or an alternative splice variant
↓
Also: cut out the spot and identify the protein that sits on it
→ the whole point of the screen
```

The comparison you must say out loud: *"LC-MS gives a mass fingerprint that has to be matched against a database, so it cannot identify a novel protein — which is exactly what protein discovery wants. LC-MS/MS adds a second fragmentation stage and reads the peptide sequence directly, so it can identify novel proteins and novel splice variants without any reference."*

Self-check. Name the two components of biomarker discovery from 2D gel. → *Protein separation (2D-PAGE) and protein identification (mass spectrometry).*

Chain 8 — Quantifying proteins and cells

The one-line idea: Gels are laborious and only semi-quantitative; if you found a soluble biomarker you need an assay a clinic can actually run.

The chain:

```
Found a soluble biomarker in blood / saliva / urine
↓
Gel is labour-intensive, time-consuming, NOT quantitative → need an assay
↓
ELISA — quantitative, for soluble protein
capture antibody binds the antigen in the sample
DIRECT: detection antibody is enzyme-conjugated (HRP) → substrate → colour → read
INDIRECT: detection antibody is NOT conjugated → enzyme-conjugated SECONDARY binds it
several secondaries per primary → SIGNAL AMPLIFICATION → more sensitive at low level
↓
MULTIPLEX ASSAY: 10–20 proteins in one sample
detection beads identify WHICH antigen; a separate fluorophore label (streptavidin-PE) gives the LEVEL
↓
ELISpot — counts CELLS, not concentration
```

```
peptide + PBMC/T cells → IFN-γ secreted → captured as a spot → count the spots
↓
FLOW CYTOMETRY — CD8 population, activation markers; IFN-γ is the most specific T-cell activation marker
(IL-2 and IL-15 are alternatives but less specific)
```

The comparison you must say out loud: *"ELISA measures the concentration of a soluble protein; ELISpot measures the number of antigen-specific cells secreting a cytokine. And the indirect ELISA is more sensitive than the direct one because several enzyme-conjugated secondary antibodies bind each primary, amplifying the signal — which matters when the biomarker is at low abundance."*

Self-check. Which ELISA format, and why, for a low-abundance circulating biomarker? → *Indirect, for the amplification.*

Chain 9 — Perturbing the genome: the CRISPR screen

The one-line idea: This is the named exam question. Understand the logic, not the apparatus.

The chain:

```
We want to find the gene that causes phenotype X (e.g. drug resistance)
↓
Gene function is studied by REMOVING it: loss-of-function screen
↓
Why it works: Cas9 cuts a double-strand break at the target
↓
The cell repairs it with NHEJ — which is ERROR-PRONE and template-free
→ indels → frameshift → gene knocked out
(this is the L4 material reused: NHEJ is the same pathway that fails in SCID)
↓
Introduce TWO things to the culture:
① the Cas9 enzyme ② an sgRNA library (whole genome, or focused on drug-metabolism genes)
gRNAs are packaged into lentivirus for delivery
↓
Drug treatment → phenotype → SURVIVING cells are sequenced (NGS) → which sgRNAs enriched/depleted?
↓
POSITIVE screen: knockout ENRICHED among survivors → gene = DRUG SENSITIVITY (gene A)
NEGATIVE screen: knockout DEPLEATED (cells died) → gene = DRUG RESISTANCE (gene B)
↓
CONTROL ARM (no drug): if knockout cells vanish anyway → gene is simply ESSENTIAL for survival (gene C)
↓
Then: target validation + pathway enrichment analysis on the genomic data
→ example: a single gene responsible for sorafenib resistance in liver cancer
```

The comparison you must say out loud: *"In a positive screen the knockout is enriched among the surviving cells, so the gene confers drug sensitivity; in a negative screen the knockout is depleted because those cells die, so the gene confers drug resistance. The untreated control arm is what separates a genuine drug-response gene from a gene that is simply essential for cell survival."*

Self-check. When would you use a focused library rather than a whole-genome library? → *When the phenotype is narrow and you already know the candidate space — e.g. genes in drug metabolism — because a focused library makes the analysis cleaner and cheaper. A whole-genome library is for phenotypes like cell differentiation where you cannot predict*

the genes.

Chain 10 — Making the protein in a cell

The one-line idea: The last chain is pure engineering, and it is where "what do you actually do in the lab" questions come from.

The chain:

```
We want our human protein, made in a mammalian cell
↓
Problem: DNA is a CHARGED molecule and the plasma membrane is a hydrophobic lipid bilayer
→ naked DNA cannot cross it
↓
Route 1 — TRANSFECTION: chemical reagents (e.g. lipofectamine) package the DNA into vesicles
which fuse with the membrane
works well for many cell lines
FAILS on primary immune cells: T cells, NK cells, macrophages
↓
Route 2 — ELECTROPORATION: a rapid voltage pulse creates TRANSIENT PORES in the membrane
DNA enters during the pore, the membrane reseals when the pulse ends
THE method for T cells, NK cells and macrophages
↓
Choice of host cell: CHO (Chinese hamster ovary) cells for making human protein
↓
CODON OPTIMISATION: CHO uses a different codon set from humans
→ rewrite the coding sequence with human→CHO codons to maximise yield
↓
Promoter drives transcription: CMV, SV40, etc.
↓
TAGS — what do you need the tag FOR?
need to detect it? → Fc fusion (or GFP)
need it to be stable? → Fc fusion (or GFP)
need it to dimerise? → Fc fusion (the Fc forms an inter-chain disulfide bond)
need to know where it is? → GFP / RFP
↓
LINKER: you cannot fuse GFP straight onto a functional protein
→ use a flexible linker, commonly Gly₃Ser
↓
Functional characterisation: GFP localisation study (e.g. HEK293T, HeLa, HepG2)
```

The comparison you must say out loud: *"Chemical transfection packages DNA in lipid vesicles and works for most cell lines, but primary T cells, NK cells and macrophages are refractory to it, so electroporation is used: a voltage pulse makes transient pores in the membrane through which the DNA enters, and the membrane reseals afterwards."*

Self-check. Your target is a nuclear receptor that is only functional as a dimer, and you have no antibody against it. Which tag, and what does it give you? → *Fc fusion: it gives you the forced homodimer (the functional requirement), plus stability, plus detection via an anti-Fc antibody.*

B2 The misconceptions to delete before you start

You think	Actually
"Formalin-fixed tissue is the standard, so it must be fine for everything."	It is the standard **for histology and IHC**. For profiling it is degraded. The two uses need different material.
"Serum and plasma are just two names for blood."	One is clot-derived, one is anticoagulated. The RA example shows they can give **opposite** results for the same analyte.
"A higher Ct means more target."	**Inverted.** High Ct = low starting copy number.
"SYBR Green is more accurate because it is newer."	SYBR Green is cheaper and easier; TaqMan is more **specific**, because only the intended product generates signal.
"RNA-Seq replaces microarray, so I do not need to know microarray."	You must know **why** it replaced it — the saturation and the novel-transcript argument is the examinable content.
"Bulk RNA-Seq gives the expression profile of the tumour."	It gives the **average** of every cell in the tumour. The tumour cells, fibroblasts and infiltrating T cells are averaged together.
"ChIP-Seq proves the binding."	It **maps** the binding. You must then validate with a reporter assay.
"LC-MS identifies proteins."	LC-MS identifies **known** proteins by fingerprint matching. Only LC-MS/MS can handle a novel one.
"The indirect ELISA is worse because it has an extra step."	The extra step **amplifies** the signal, which is what you need at low abundance.
"A positive CRISPR screen finds the gene that survives."	A positive screen finds the gene whose knockout **survives**, i.e. the gene that confers **drug sensitivity**. The gene whose knockout **dies** is the drug-**resistance** gene, found by a negative screen.
"Transfection and electroporation are interchangeable."	Not for primary T cells, NK cells or macrophages.
"A GFP tag is just for detection."	GFP is mainly for **localisation and co-localisation**; Fc is the one that forces **function** (dimerisation).

B3 A five-day plan from zero

Each day is 2–2.5 hours of focused work. Do them in order: the chains are ordered so that each one supplies the vocabulary the next one needs.

Day	Chains	Focus	Output you must have
1	B0, Chain 1	Sample quality and logistics	You can explain GIGO, FFPE vs FF, serum vs plasma, and why a central lab exists — in five sentences, in English.
2	Chain 2, Chain 3	PCR and the three transcriptomics	You can draw the RT→PCR→Ct pipeline and state the three questions microarray / RNA-Seq / scRNA-Seq each answer.
3	Chain 4, Chain 5	ChIP-Seq, clonal expansion, cancer vaccines	You can walk the TCR → neoantigen → vaccine chain and say where validation is mandatory.
4	Chain 6, Chain 7, Chain 8	Proteomics: separate, identify, quantify	You can state the separation axis of each technique and the one thing LC-MS cannot do.
5	Chain 9, Chain 10, then Part A	The named question, then revision	You can define positive vs negative screen with the control arm; then drill Part A.

The two drills that matter more than the reading.

1	**Say the discriminator aloud.** Once per day, pick five rows from the A4 table and say each one as a single English sentence. If you hesitate, that row is not learned. This is what he meant by reading primary literature to improve your English.
2	**Write one short answer per day.** Pick a question from A8, answer it in English in four or five sentences using the four moves, then mark it against the key. Do not read the key first.

Day 5 evening, and the night before any exam: read Part A only. Read A2 (the traps) and A4 (the pairs) and A3 (the numbers) and stop. Do not start a new chain the night before the paper.

Figure L5 — The five-day plan from zero

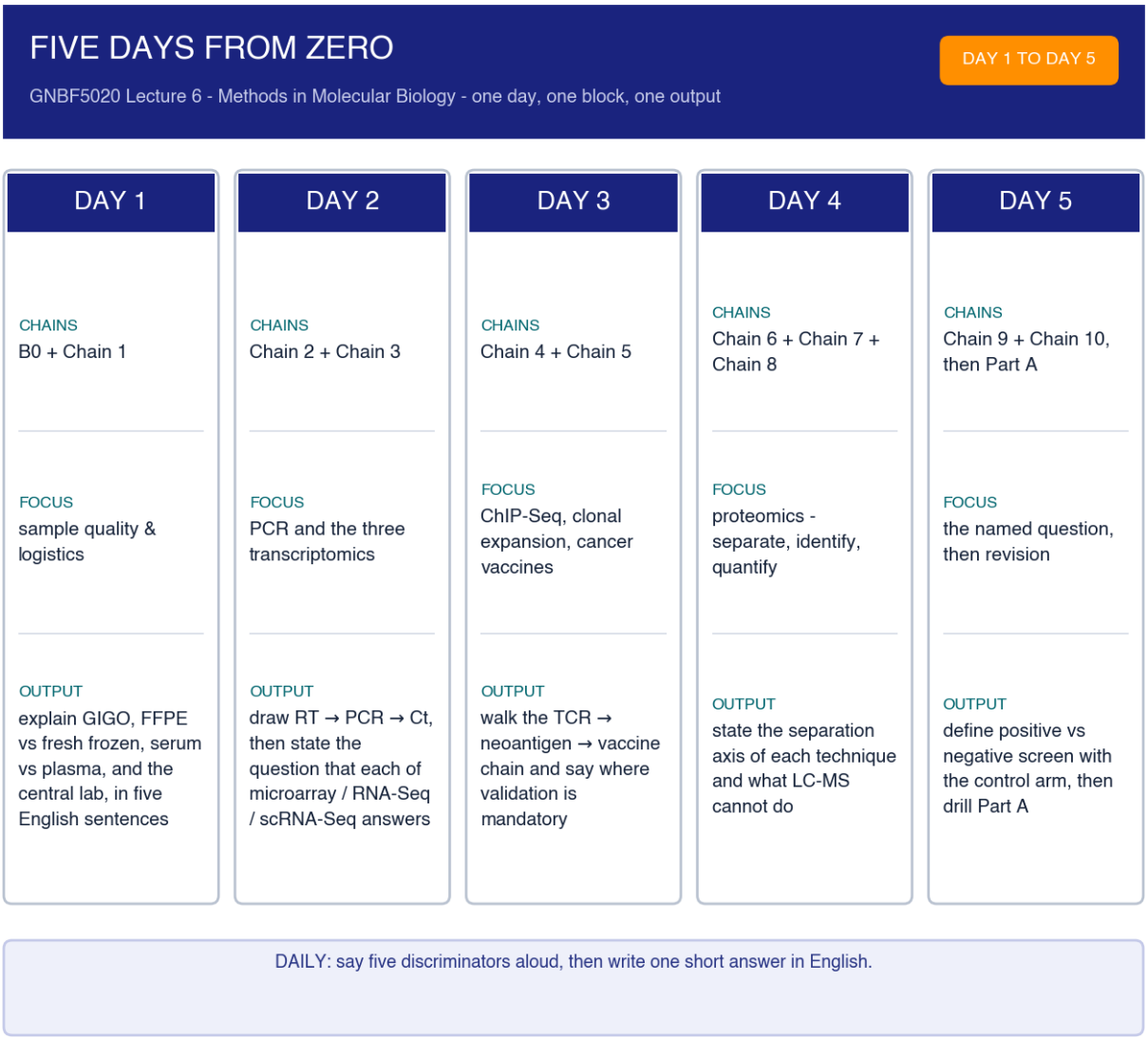


Figure L5. The five days, the chains that belong to each day, and the output you must have at the end of it, with the two drills that matter more than the reading. Figure authored at 6.5 in print width and placed 1:1 on this page.

How to read this plate: one row per day, in order - the chains are ordered so that each supplies the vocabulary the next one needs. Tick the right-hand column as you go. On day 5 evening, and on the night before any exam, read Part A only.

B4 On the day

- **Answer in English.** Zero marks otherwise. Write complete sentences, not fragments.
- **Read the verb.** "Explain" needs a mechanism. "Describe" does not. Getting this wrong was the single biggest problem in the midterm.
- **Put the number in.** If the question gives you a number, use it in your answer.
- **Name the molecule.** Never write "the target" — name the gene or protein.
- **Close with a limit.** One sentence on what the conclusion cannot be generalised to. This is cheap and it is what separates a strong answer.
- **Do not over-write.** He criticised a "very lengthy, but not so focused" answer. Short and clear beats long and thorough.

Appendix A — Where each claim in this book comes from

Everything here is traceable to the Lecture 6 recording of 7 Nov 2026 or its 46-slide deck. The claims that are **the lecturer's own words** are marked with a quotation; everything else is a faithful restatement of what he taught.

Section	Source
A1, A2	Direct quotation from the lecture, segment 1 (opening, midterm review) and segment 6 (CRISPR screening)
A3.1	Midterm statistics, trial logistics, and city list — segment 1 and 2
A3.2	Bioanalyzer, rRNA, RIN — segments 1–2; slides 3–4
A3.3	Real-time PCR, Ct, SYBR vs TaqMan — segment 2; slides 7–10
A3.4	Microarray vs RNA-Seq numbers — segment 3; slides 19–21 (Rao et al., *Frontiers in Genetics* 2019)
A3.5	SDS-PAGE, IgG, IEF, 2D-PAGE, MS — segment 5; slides 29–33
A3.6	ELISA, multiplex, ELISpot, peptides, Fc, transfection, electroporation, Gly_3Ser — segments 5–6; slides 34–45
A4	All seventeen pairs are drawn from the comparisons he made explicitly in lecture
A5	Method selection, built from his stated reasons for each technique
A6	The four moves and the habits come from his midterm feedback and the "explain, don't describe" correction
A7	His explicit "Don't worry about all this details" and "No need to memorize it" statements, segments 2–4
B1	Chain-by-chain reconstruction of the lecture in teaching order; each chain's "comparison" sentence is the form he uses when he compares two techniques aloud

Appendix B — Key figures from the lecture deck

Slides	Topic
2	Overview — methods in molecular biology: samples, genomics, proteomics, cell biology
3–6	Biological samples: GIGO, FFPE vs FF, serum vs plasma, the sample-logistics journey, trial-supply market
7–10	Real-time PCR: principle, SYBR vs TaqMan, workflow, multiplex, Ct normalisation
11, 13	Microarray: principle, two-colour vs one-colour workflow
14–17	NGS platforms, ChIP-Seq, RNA-Seq
18–21	RNA-Seq workflow, and the microarray vs RNA-Seq comparison with Table 1A / 1C
22–24	scRNA-Seq: principle, microfluidics library construction, workflow
25–26	Multi-tissue T-cell transcriptomics; the pan-cancer tumour-infiltrating T-cell atlas
27–28	Clonal expansion and TCR sequencing for the personalised cancer vaccine
29–33	SDS-PAGE, IgG, isoelectric focusing, 2D-PAGE, mass spectrometry
34–36	Antigen identification by MS, ELISA formats, multiplex assay
37–38	CRISPR/Cas9 library screening, positive vs negative screen, control arm
39–40	Protein transfection; expression of proteins by mammalian cells
41–45	Chemical transfection vs electroporation; Fc fusion and GFP tags

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Appendix C — External verification, and the corrections it produced

Everything in Parts A and B is drawn from the lecture recording. This appendix records the independent check of the technical claims against published sources, because two of the lecturer's slide claims are **simplified** and one of them is **materially different from the paper he cites**. If a question ever asks you to go beyond the slide, this is the page to read.

C.1 The course

Item	Verified	Source
Institution	**The Chinese University of Hong Kong (CUHK)**, School of Biomedical Sciences, Faculty of Medicine	http://bioinfo.cuhk.edu.hk/programmes.php
Programme	MSc in Genomics and Bioinformatics; GNBF5020 *Introduction to Molecular Biology and Genetics* is a **3-unit** course in Term 1	http://bioinfo.cuhk.edu.hk/programmes.php · https://www3.sbs.cuhk.edu.hk/en/msc-in-genomics-and-bioinformatics
Class time	**Saturday 16:15–18:30**, CUHK campus	same
Medium of instruction	**English** (course is tagged (EN) in the CUHK registry)	same
Assessment weighting	**Assignments 30 % · mid-term 30 % · final 40 %**	https://www.wfsfaa.gov.hk/tc/resources/course_search/cef_details.php?course_code=41M124753
Past paper / sample exam	**None publicly indexed.** No exam blueprint or specimen paper is published for this course.	—

What this means for you. The paper structure you experience (MCQ + short answer) is set by the lecturer, not published — but the published **30 % mid-term** component is consistent with what he described, and the **40 % final** tells you the final paper carries the largest single share of the course grade. That is a reason to treat the final seriously, and it is the one external fact about the exam that is actually documented.

C.2 Corrections you should know about

Slide claim	What the sources say	What to do
The slide cites *"Rao MS et al., 2019"* with **no journal named**	The paper is **Rao MS, Van Vleet TR, Ciurlionis R, Buck WR, Mittelstadt SW, Blomme EAG, Liguori MJ (2019). *Comparison of RNA-Seq and Microarray Gene Expression Platforms for the Toxicogenomic Evaluation of Liver From Short-Term Rat Toxicity Studies*. Frontiers in Genetics 9:636. doi:10.3389/fgene.2018.00636** (PMID 30723492).	Cite *Frontiers in Genetics* 2019 if a citation is asked for.
The slide's Table 1A / 1C numbers	**Checked against the published paper: they match exactly.** Table 1A — ANIT 1014 (array) / 3139 (RNA-Seq), APAP 92 / 512, CCl_4 689 / 2127, DCLF 16 / 196, MDA 1081 / 3672. Table 1C — ANIT 41 / 577.5, APAP 20.3 / 363.8, CCl_4 12.9 / 143.8, DCLF 2.1 / 50.9, MDA 15.7 / 226.5. The paper additionally reports Spearman's ρ = 0.30 for DCLF (and 0.44 for APAP); the slide's scatter plots only four treatments, so DCLF has no R there.	**Quote his slide.** It is accurate to the paper.
Slide cut-off: **RIN < 6 = poor quality** (its own examples: RIN 10 intact, RIN 5 partially degraded, RIN 3 strongly degraded)	Agilent's official scale is **1–10 with 10 = cleanest** (app note 5989-1165EN) — the slide **agrees** with it. One practical addition: for FFPE, RIN is less informative because the ribosomal peaks are lost, so **DV200** (>200 nt fraction) is the recommended FFPE metric.	Learn the cut-off **RIN < 6 = poor quality**.
Ct ~13–17 for house-keeping genes	Correct for the *most abundant* reference genes (β-actin 13.93–18.63; 18S from 14.86). A six-gene panel spans **13.93–26.09**.	Use 13–17 as he taught it; know the wider band.
rRNA > 95 % of total cellular RNA	The slide's figure. The most commonly quoted share of total RNA *mass* that is rRNA is **~80–85 %**.	Cite the slide's number in the exam; it is what he taught.

C.3 Claims that checked out exactly as taught

These were verified and need no adjustment — you can state them with confidence:

- **TaqMan probe.** 5′ reporter + 3′ quencher; the **5′→3′ exonuclease activity of Taq** cleaves the reporter off during extension, releasing the signal. The slide's model is correct.
- **SYBR Green.** Binds the minor groove of **any** double-stranded DNA and is therefore **not sequence-specific** — it detects non-specific products and primer-dimers, which is exactly why it loses specificity. (https://www.agilent.com/cs/library/applications/5989-1165EN.pdf · Agilent/ Bio-Rad dye documentation)
- **Ct is inversely related to starting copy number.** Relative expression is computed from ΔCt = Ct(target) − Ct(reference) and the 2^−ΔΔCt method.
- **IgG on SDS-PAGE.** Intact ≈ **150 kDa** non-reducing; reducing with DTT or β-mercaptoethanol gives **~50 kDa heavy** and **~25 kDa light** chains.
- **Isoelectric focusing.** Proteins migrate to the pH at which their **net charge is zero** and stop there; above the pI the protein is net negative, below it net positive. **2D-PAGE: first dimension pI, second dimension molecular weight** — confirmed.
- **LC-MS / LC-MS/MS.** Peptide mass fingerprinting is **database-driven** and cannot identify a protein absent from the database (so not a novel protein or a novel splice isoform); tandem MS fragments the peptide to **b- and y-ion series** and performs **de novo sequencing** without any reference.
- **2D-DIGE.** Two pools labelled with **Cy3 and Cy5 NHS esters**, co-migrated on one gel; overlay semantics are **channel-relative** — red = higher in the Cy5 channel, green = higher in the Cy3 channel, yellow = equal (within ~2-fold). Always state which dye carries which sample.
- **CRISPR knockout mechanism.** Cas9 cleavage is repaired by **error-prone NHEJ**, introducing indels and especially **frameshifts** that disrupt gene function — the same NHEJ you met in Lecture 4.
- **Anti-CD3 / anti-CD28.** Anti-CD3 delivers the **TCR signal (signal 1)**; anti-CD28 delivers the **co-stimulatory signal (signal 2)**.
- **Transfection vs electroporation.** Lipid transfection is inefficient on primary immune cells; the Neon electroporation system is specifically recommended for primary T, NK and macrophage cells.
- **10x Genomics scRNA-Seq.** Gel Bead-in-Emulsion droplets with **barcoded gel beads carrying UMIs**; **3′** chemistry captures via poly(dT) on the gel-bead primer with a **template-switch oligo**, and the **5′** chemistry supplies poly(dT) as

the RT primer — so do not present poly(A) capture and TSO as one single mechanism.

- **MammaPrint.** A **70-gene** prognostic signature derived from ~25,000 genes on an **Agilent Hu25K microarray** (231 regulated genes narrowed to 70), now commercialised by **Agendia**, **FDA-cleared** (cleared, not approved), reporting **high-risk / low-risk** for distant recurrence. For contrast, the 21-gene **Oncotype DX** RT-PCR assay is the one incorporated into ASCO and NCCN guidelines.
- **Codon optimisation in CHO.** Correct in principle, but note that mammalian cells tolerate broader codon usage than *E. coli*, so the effect is smaller than in bacteria; keep GC around 40–55 % and do not treat a high CAI as a guarantee of expression.

C.4 Where the literature and the lecture use different words for the same thing

The CRISPR screen naming is the one place where an answer can lose a mark by using the wrong synonym. The standard vocabulary is:

Readout	Standard names	Lecturer's names	Same thing?
Knockout **enriched** among survivors	positive selection · FOF (fitness-of-fitness)	**positive screen**	[yes] yes
Knockout **depleted** — the cells die	negative selection · dropout	**negative screen**	[yes] yes

His convention matches the standard one, so you may use either vocabulary. The mark is lost by *mis-describing the readout*, not by choosing a different name. Always lead with the readout — "the knockout is enriched / depleted" — and then name the gene's consequence.

C.5 What could not be verified, and is therefore not claimed in this book

- No published **past paper, sample exam or exam blueprint** for GNBF5020 is publicly available. Every prediction in Part A is therefore drawn from the lecturer's own statements in the recording, not from a published paper.
- The **paper structure** — he described the midterm as an MCQ section plus short questions — is the lecturer's description of what he sets; no public document specifies it.
- The current **guideline status of MammaPrint** in 2026 was not confirmed beyond the sources cited above; only its 70-gene composition, microarray origin and FDA clearance are stated here.
- No quantified head-to-head **serum vs plasma** figure was retrieved; only the pre-analytical principle and the lecturer's own rheumatoid-arthritis example are used.

Verified 31 Oct 2026 against the sources listed above. Where the lecture and the literature differ, this book says so explicitly — because "seeing is believing" is his instruction to you, and the best way to honour it is to know which number he will want on the page and which one is actually right.